

Matter Carriers: Quarks, Leptons, and Neutrinos

Spinor Skins, Mass Labels, Flavor Routes, and No Primitive Matter-Identity Escape in the Standard Model

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Abstract

This paper is SM-3 in the Standard Model carrier-exhaustion sequence. It imports the SM-1 master classifier and the SM-2 gauge-sector closure, then applies that machinery to matter standing. The target is not to replace Standard Model matter phenomenology, derive numerical quark masses, derive exact charged-lepton pole masses, determine absolute neutrino masses, compute PMNS entries, classify every named BSM model, or prove the global Standard Model endpoint. The target is scoped primitive-status closure for the matter sector under a fixed family-cardinality envelope.

Quark, charged-lepton, and neutrino standing is not primitive by spinor notation, field label, representation label, chiral label, mass value, Yukawa coordinate, flavor name, generation label, detector track, hadron membership, parton route, oscillation baseline, cosmological bound, PMNS coordinate, EFT coefficient, or isolated-particle skin. These remain lawful representatives, witnesses, routes, projections, transport ledgers, mixing statuses, extension dispositions, target updates, burdens, or failed primitive promotions. Primitive matter standing is carrier-indexed through representation response, color-bearing matter response, colorless charged matter response, neutral weak matter response, and licensed mass-response-slot standing. Flavor and oscillation language is preserved as a route/interface status rather than inserted into the primitive neutrino carrier.

The closed matter carrier package is

$$C_{\text{matter}} = \{C_{\text{matter-rep}}, C_{\text{color-bearing-matter}}, C_{\text{colorless-charged-matter}}, \\ C_{\text{neutral-weak-matter}}, C_{\text{mass-response-slot}}\}.$$

Using the SM-1 classifier, the SM-2 gauge/color/weak handoff, the standalone Standard Model structural spine, realized physical interior, representation-response and representation-algebra support, charge-lattice support, Yukawa/mass equivalence, anomaly/Yukawa compatibility, charged-lepton residue support, neutrino reconstruction support, and observable-transport support, the paper proves the scoped matter-sector deletion result

$$\text{Survprim}_{D_{\text{matter}}^{N_g}} (\text{Attr}(D_{\text{matter}}^{N_g}) \setminus C_{\text{matter}}) = 0.$$

This result is primitive-status closure under a fixed same-scope matter arena. It preserves standard field, particle, mass, flavor, detector, oscillation, EFT, and BSM language as licensed downstream language.

The AASC/kernel dependency used here is not an overlay on matter phenomenology. It is the minimal non-degeneracy condition under which matter-sector claims preserve standing, reference,

sameness, comparison, prediction, and falsification. Detector tracks, oscillation routes, hadron membership, mass values, and family labels are therefore not demoted by convention; they are classified by the role they can non-degenerately occupy inside the same-scope matter arena.

Keywords: AASC; Standard Model; matter carriers; quarks; charged leptons; neutrinos; representation response; spinor fields; Yukawa bookkeeping; mass-response slot; neutrino oscillation; flavor route; confinement-compatible quark standing; derivative skin; primitive closure; BSM audit.

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1 Kernel Status: Non-Degeneracy, Not Overlay

The AASC kernel is used here in its upstream role. It is not an interpretive overlay placed on physics, and it is not a discretionary taxonomy imposed after the physical content is fixed. It states the minimal non-degeneracy conditions under which a physical claim can retain standing, reference, sameness, comparison, prediction, and falsification at all. Without such conditions, the expressions “same claim”, “same target”, “same regime”, “same comparison”, “same prediction”, and “same failure” lose determinate force.

For this reason, SM-3 is not a vocabulary substitution for ordinary physics. It is a local matter-sector primitive-status closure theorem over kernel-preserved claimhood. The familiar objects in the matter arena may be physically real, measurable, predictive, and indispensable in calculation. That does not by itself make them primitive carriers. Primitive standing requires a same-scope role that remains standing-bearing under the admissible transformations of the fixed target.

Definition 1.1 (Non-degenerate matter claim). *A matter-sector claim is non-degenerate only when its target, reference-bearing status, admissible comparison class, and standing conditions are preserved. In notation,*

$$\text{NonDegenerateClaim}_{\text{matter}}(x) \Rightarrow \text{KernelPreserved}_{\text{matter}}(x).$$

The implication is not a convention of this paper; it records the necessity condition for the claim to remain an identifiable object of assertion, prediction, comparison, or failure.

Theorem 1.2 (Sameness requires kernel preservation). *For SM-3 claims in $D_{\text{matter}}^{N_g}$, sameness is kernel-preserved sameness:*

$$\text{SameClaim}_{\text{matter}}(x, y) \Rightarrow \text{KernelPreserved}_{\text{matter}}(x, y).$$

Consequently,

$$\neg \text{KernelPreserved}_{\text{matter}}(x, y) \Rightarrow \neg \text{SameClaim}_{\text{matter}}(x, y).$$

A non-preserved target, comparison class, admissibility rule, or reference-bearing status yields a replacement, degeneration, target update, or failed comparison rather than the same claim under test.

Theorem 1.3 (Prediction and falsification require non-degenerate reference). *Every prediction or falsification claim in the matter arena presupposes preserved target identity, comparison class, and reference standing:*

$$\begin{aligned} & \text{Prediction}_{\text{matter}}(x) \vee \text{Falsification}_{\text{matter}}(x) \\ & \Rightarrow \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x) \wedge \text{ReferenceStanding}(x). \end{aligned}$$

A prediction is a prediction of the same claim only under sameness preservation. A falsification is a falsification of the same claim only under target-preserving failure.

Proposition 1.4 (Kernel-denial burden). *A denial of the kernel is not a successful rejection of SM-3 unless it recovers the functions that the kernel supplies. Formally, any denial must satisfy the burden*

$$\begin{aligned} \text{KernelDenied}(x) \Rightarrow & \text{RecoverSameness}(x) \wedge \text{RecoverStanding}(x) \\ & \wedge \text{RecoverReference}(x) \wedge \text{RecoverFalsification}(x). \end{aligned}$$

Thus a matter counterexample must preserve the same target, same comparison class, and same reference-bearing object while showing that representation-response, color-bearing matter, colorless charged matter, neutral weak matter, and mass-response-slot standing can be given without the kernel conditions used here.

Remark 1.5 (Proof class). *The result is not a Lagrangian reconstruction, not a numerical parameter derivation, not an empirical-finality theorem, and not a no-new-physics theorem. It is a same-scope primitive-status exhaustion theorem for the stated arena. New measurements, anomalies, effective descriptions, and target-changing discoveries remain admissible; they do not become unlicensed same-scope primitive carriers merely by being real or useful.*

Reading roadmap

SM-3 proceeds in six stages. Sections 2–4 fix the claim boundary, dependency posture, theorem status, and matter-sector arena. Sections 5–8 define primitive matter predicates, matter carrier certificates, matter skin identification, and carrier-by-carrier certificate discharge. Sections 9–13 classify representation response, quarks, charged leptons, and neutrinos. Sections 14–16 handle mass-response/Yukawa bookkeeping, generation labels, detector routes, oscillation routes, cosmological bounds, and projection statuses. Sections 17–19 provide BSM/EFT branch certificates, the matter prediction box, scoped closure theorem, and hostile reconstruction. The appendices provide source ledgers, carrier registers, skin atlases, certificates, theorem inventory, standard-physics orientation, formalization target, and the SM-6 handoff package.

2 Reader orientation: what SM-3 proves

2.1 One-sentence thesis

Matter particles are not primitive because they are spinor fields, flavor labels, mass values, detector tracks, or oscillation routes. Their standing is carrier-indexed through representation-response, color-bearing matter response, colorless charged response, neutral weak response, and licensed mass-response slots. Flavor and oscillation are preserved as boundary, route, mixing, or reconstruction statuses; they are not promoted into primitive matter standing by route availability alone.

This sentence is a primitive-status claim, not a rejection of Standard Model matter theory. The ordinary language of fermion fields, quarks, leptons, neutrinos, tracks, jets, decays, oscillations, masses, Yukawa couplings, and flavor matrices remains available. What is denied is the automatic inference from an available display to primitive matter standing.

2.2 SM-3 in the six-paper sequence

SM-1 constructed the master classifier, the no-promotion theorem for derivative skins, target-update separation, burden status, non-ad-hoc classifiability, and the prediction-register grammar [2]. SM-2 applied that classifier to the gauge sector and closed the local gauge carrier package

$$C_{\text{gauge}} = \{C_{\text{gauge-order}}, C_{\text{transport}}, C_{\text{curv/hol}}, C_{\text{charge-response}}, \\ C_{\text{color-transport}}, C_{\text{weak-chiral}}, C_{\text{anomaly}}, C_{\text{gauge-vacuum}}\}$$

with the scoped theorem

$$\text{Survprim}_{D_{\text{gauge}}}(\text{Attr}_G \setminus C_{\text{gauge}}) = 0$$

inside the fixed gauge arena [3]. SM-3 receives those outputs rather than rederiving them. Its local endpoint is the matter analogue

$$\text{Survprim}_{D_{\text{matter}}^{N_g}}(\text{Attr}(D_{\text{matter}}^{N_g}) \setminus C_{\text{matter}}) = 0.$$

The global Standard Model endpoint remains the SM-6 synthesis target.

2.3 Standard usage preserved

SM-3 does not deny quark fields, charged-lepton fields, neutrino fields, spinor notation, Dirac/Weyl/Majorana language, current masses, constituent masses, pole masses, tracks, jets, hadrons, oscillations, cosmological neutrino bounds, Yukawa parameters, mass matrices, PMNS language, EFT operators, or BSM matter models. It classifies them. The paper asks what status such language can carry under the fixed same-scope matter arena.

2.4 Primitive-status claim

The denied inference is

$$\text{spinor/mass/fluxor/track/route/projection} \Rightarrow \text{primitive matter standing.}$$

The positive claim is

$$\text{matter standing} \Rightarrow \text{carrier-indexed matter-response role.}$$

The distinction is important because many matter claims arrive through surfaces that are valid but nonprimitive: a field symbol, a mass value, a flavor tag, a detector pattern, an oscillation baseline, a hadron route, or an EFT coefficient.

Nonclaim 2.1 (What SM-3 does not claim). *SM-3 does not claim exact quark mass derivation, exact charged-lepton pole-mass closure, exact neutrino mass derivation, PMNS numerical closure, CKM numerical closure, full flavor/mixing/generation closure, derivation of all Yukawa eigenvalues, proof that sterile-neutrino models are false, proof that vector-like fermions are false, or global Standard Model endpoint closure. Generation-cardinality-changing proposals are classified in SM-3 as burdens, failed primitive duplications, or target updates unless a role-cardinality certificate is supplied; full generation-cardinality closure is handed to SM-5 and SM-6.*

Status 2.2 (What SM-3 claims). *SM-3 claims that, in the fixed matter-sector arena $D_{\text{matter}}^{N_g}$, primitive matter standing is exhausted by the closed matter carrier package plus registered derivative statuses. Spinor fields, mass labels, flavor labels, detector tracks, oscillation routes, hadron membership, Yukawa entries, and BSM matter labels do not self-promote into primitive matter standing.*

3 Dependency ledger and proof-class boundary

3.1 Upstream kernel dependency and source direction

The dependency order is fixed as follows. The kernel/admissibility layer supplies the minimal non-degeneracy conditions for sameness, standing, reference, comparison, prediction, and falsification. SM-3 instantiates those conditions in the matter arena; it does not derive the kernel from the Standard Model or from any later structural-spine paper. Later Standard Model structural results may instantiate the same kernel at larger scope, but they do not supply the upstream necessity of kernel preservation.

Accordingly, dependence on the kernel is not a defect in the proof class. A successful objection must either break a received local closure premise, supply a faithful same-scope primitive certificate, or recover non-degenerate physical claimhood without the kernel functions listed in Proposition 1.

3.2 Imported SM-1 machinery

SM-3 uses the SM-1 component normal form

$$R_x = (D_{\text{SM}}, x, C_x, \text{Role}_x, \text{Rep}_x, \text{Meas}_x, \text{Transport}_x, \text{Route}_x, \\ \text{Mix}_x, \text{Proj}_x, \text{Extension}_x, \text{Update}_x, \text{Cert}_x, \rho_x, \text{Pred}_x).$$

The imported classifier results are: carrier-admission conditions, skin-identification tests, no primitive skin promotion, target-update discipline, counterexample burden as a review state rather than survivor status, non-ad-hoc classifiability, the prediction-register format, and the falsification protocol [2].

3.3 Imported SM-2 machinery

SM-2 contributes the closed gauge, color, charge, weak, anomaly, and gauge-vacuum roles used by matter claims. The most important handoff items are color transport for quark standing, charge response for charged matter, weak-chiral response for weak matter, gauge-vacuum compatibility for weak response, anomaly obstruction for representation survival, and the gauge side of no isolated color skin [3]. SM-3 treats these as local inputs and closes the matter side.

3.4 Matter dependency package

The matter proof uses the following compressed dependency ledger. The table records the imported output, the proof function inside SM-3, and the nonclaim boundary.

Table 1: SM-3 dependency ledger

Source layer	Imported output	Proof use	Nonclaim boundary
SM-1 classifier	Claim arena, carrier/skin grammar, target update, burden, prediction register	Status machine for all matter claims	Does not prove local matter closure alone [2].
SM-2 gauge carriers	Closed gauge package, including color transport, charge response, weak-chiral response, anomaly, and gauge-vacuum roles	Gauge-side carriers for quark, charged-lepton, and neutrino claims	Does not prove matter-side closure alone [3].
Standalone SM spine	Structural tuple with gauge product, representation structure, Higgs bridge/fixation, Yukawa ledger, charge ledger, anomaly ledger, vacuum form, mass slot, mixing slot, and family envelope	Structural arena and parameter boundary	Does not derive the empirical parameter table [4].
Realized physical interior	Fields as persistence carriers; particles as localized persistence classes; mass, vacuum, gauge, causality stack	Physical grounding of field and particle states	Does not make field symbols primitive [5].

Source layer	Imported output	Proof use	Nonclaim bound-ary
Representation admis-sibility	Pre-linear representation response and charge-witness ledgers	Positive carrier sup-port for $C_{\text{matter-rep}}$	Does not assume vector spaces or representation theory [6].
Representation algebra specialization	Algebraic representation spaces as witness-preserving realizations	Negative no-promotion for spinor, chiral, tensor, dimension, and multiplicity skins	Does not derive SM representa-tion table [7].
Representation-response rigidity	Compression of representation-response roles	Positive carrier sup-port for response discipline	Does not close generation cardi-nality by itself [8].
Charge lattice and L1 charge	Charge-lattice and hypercharge/electric-charge normalization discipline	Positive support for charged and neutral predicates	Does not derive gauge couplings [9, 10].
L1 anomaly/Yukawa support	Compatibility between anomaly discipline and Yukawa realizability	Chiral matter and BSM guard	Does not derive Yukawa eigenval-ues [11].
L1 Yukawa/mass equivalence	Cross-skin collapse between Yukawa and mass bookkeep-ing	Negative over-count collapse for mass/Yukawa skins	Does not close full mass-slot standing alone [12].
Yukawa compatibility	Parameter-survival classifica-tion over the canonical bridge carrier	Positive support for mass-response-slot standing	Does not derive mass values [13].
Generation-count sup-port	Residual-exhaustion theorem for fixed-boundary generation data	Fixed family-cardinality envelope and duplication guard	Does not close arbitrary genera-tion alteration in-side SM-3 [14].
Charged-lepton sup-port	Physical residue, role-projection readout, and readout closure for charged-lepton mass-shape status	Charged-lepton mass-shape and readout skin discipline	Does not make SM-3 a numeri-cal lepton paper [15, 16, 17].
Neutrino reconstruc-tion and observable transport	Flavor reconstruction under noncommuting propagation, branch orientation, matter re-sponse, and transport-readout discipline	Neutrino route, pro-jection, and burden discipline	Does not close PMNS or abso-lute mass scale [18, 20, 22, 23, 24, 25, 26, 27, 28].

3.5 Closure proof spine

The matter-sector primitive-status closure route is

$$\begin{aligned}
& SM1 + SM2 + \Sigma_{SM} + RPI + A4 + PSM3 + SR3 + Q_{SM} \\
& + L1_{\text{charge}} + L1_{\text{anom/Yuk}} + L1_{\text{Yuk/Mass}} \\
& + MGN4 + GenCount + CL_{\text{arc}} + \nu_{\text{osc}} + AOT_{\nu} \Rightarrow \text{ClosedMCert}(C_{\text{matter}}).
\end{aligned}$$

The proof spine is decompressed into carrier-by-carrier certificates in Section 8. The final local deletion is obtained by the certificate-to-deletion lemma of Section 19.

3.6 Theorem-status ladder

Throughout SM-3, *closed* means closed for primitive-status classification under the declared same-scope matter arena. It does not mean empirical finality, numerical parameter closure, or model-by-model BSM adjudication.

Status	Meaning in SM-3
Definition	A stipulated object in the matter-sector status grammar.
Criterion	A rule for admitting or rejecting a matter-sector status assignment.
Imported theorem	A result received from SM-1, SM-2, the standalone spine, representation, charge, Yukawa, charged-lepton, or neutrino support.
Closed SM-3 theorem	A local primitive-status result produced by applying the imported machinery to $D_{\text{matter}}^{N_g}$.
Sector-local corollary	A result handed forward to SM-4, SM-5, or SM-6.
Open burden	A live matter-sector challenge lacking a closed same-scope carrier certificate.
SM-6 endpoint target	Global Standard Model primitive-status deletion, not asserted here.

Principle 3.1 (No premature global closure). *SM-3 is not SM-6. It closes the matter-sector primitive-status question under $D_{\text{matter}}^{N_g}$. The theorem*

$$\text{Survprim}_{D_{\text{SM}}}(\text{Attr}_{SM} \setminus C_{SM}) = 0$$

belongs to the final synthesis only after gauge, matter, charge/coupling/RG, flavor/mixing/generation, Higgs, BSM/EFT, and projection-interface closures are assembled.

4 The matter-sector claim arena

Definition 4.1 (Matter-sector arena with fixed family envelope). *The matter-sector claim arena is*

$$D_{\text{matter}}^{N_g} = (D_{\text{SM}}, R_{SM}, C_{\text{gauge}}, C_{\text{color}}, C_{\text{charge}}, C_{\text{weak}}, H_{\text{br/fix}}, M_{\text{slot}}, \Theta_{\text{mix}}, N_g, \text{MatterAdm}).$$

It is the subarena of the fixed Standard Model comparison domain in which matter claims are evaluated while the family-cardinality envelope N_g is held fixed.

Here R_{SM} is the representation-response structure, C_{color} is the SM-2 color-transport carrier, C_{charge} is charge-response standing, C_{weak} is weak-chiral response, $H_{\text{br/fix}}$ is the Higgs bridge/fixation carrier, M_{slot} is the mass-response slot ledger, Θ_{mix} is the mixing/phase boundary, N_g is the family-cardinality envelope, and MatterAdm records the admissibility discipline for matter standing. A generation-cardinality-changing proposal is not silently treated as same-scope; it must either preserve the fixed envelope through a role-cardinality certificate or be classified as burden, failure, or target update.

Definition 4.2 (Matter claim). *MatterClaim(x) means that x is a submitted claim about a quark, charged lepton, neutrino, spinor field, representation label, chiral label, mass value, Yukawa coordinate, generation label, flavor label, detector track, hadron membership, parton route, oscillation route, cosmological bound, sterile-neutrino surface, vector-like fermion, fourth generation, leptoquark, dark-sector fermion, matter-sector EFT operator, or matter-sector BSM extension.*

Definition 4.3 (Same-scope matter claim). *SameScope $_{D_{\text{matter}}^{N_g}}$ (x) holds when x preserves D_{SM} , the representation-response structure, gauge carrier package, charge lattice, color-transport discipline, weak-chiral response, anomaly ledger, vacuum compatibility, Higgs bridge/fixation boundary, mass-response slot discipline, family-cardinality envelope, and no-hidden-load condition.*

Criterion 4.4 (Matter target update). *A matter claim is a target update only if it changes carrier inventory, representation-family ledger, charge lattice, color-sector regime, weak-sector regime, anomaly/vacuum structure, Higgs/mass-fixation boundary, flavor/mixing domain, source/measurement domain, hidden-sector inventory, family-cardinality envelope, or the admissibility boundary.*

This criterion prevents an equivocation that is especially common in matter-sector BSM language. A new fermion label, new anomaly, new mass fit, new oscillation route, or new detector pattern is not automatically a target update. It first enters the classifier. It becomes a target update only when a target-changing condition is independently stated.

5 Primitive matter carriers and classifier predicates

Definition 5.1 (Color-bearing matter carrier).

$$C_{\text{color-bearing-matter}}(x) := C_{\text{matter-rep}}(x) \wedge C_{\text{color-transport}}(x) \wedge \text{ConfCompat}(x).$$

$\text{ConfCompat}(x)$ is a classifier discipline, not an additional empirical postulate. It means that color-bearing matter standing is evaluated under the fixed SM-2 color-transport carrier and the matter-response carrier, while external isolated-particle display, hadron membership, parton-distribution language, jet association, and scattering route receive derivative route or projection status unless a same-scope free-color matter carrier certificate closes.

Remark 5.2 (What confinement-compatible discipline is not). *ConfCompat is not the empirical confinement proof and it does not by itself close quark standing. The no-free-quark primitive-status result follows from the intersection of matter-representation response, the SM-2 color-transport carrier, the same-scope requirement, and descent exclusion. A free-color primitive would have to defeat both color-transport descent and matter-response descent while preserving $D_{\text{matter}}^{N_g}$. The color-transport carrier is received from SM-2; SM-3 adds and closes the matter-response side.*

Definition 5.3 (Color-null classifier predicate). $\text{ColorNull}(x)$ means $\neg \text{Load}_{\text{color}}(x)$ inside fixed $D_{\text{matter}}^{N_g}$: the submitted matter claim carries no admitted color-transport load. It is a null-load filter that restricts which already admitted matter-response branch is active. Colorlessness is therefore not an additional primitive substance and not a carrier component by itself.

Definition 5.4 (Colorless charged matter carrier).

$$\begin{aligned} C_{\text{colorless-charged-matter}}(x) := & C_{\text{matter-rep}}(x) \wedge \text{ColorNull}(x) \wedge C_{\text{charge-response}}(x) \\ & \wedge C_{\text{weak-chiral}}(x) \wedge C_{\text{mass-response-slot}}(x). \end{aligned}$$

The predicates ColorNull and ChargeNeutral restrict already-admitted matter-response branches; they do not add primitive carrier content.

Definition 5.5 (Charge-neutral classifier predicate). $\text{ChargeNeutral}(x)$ means zero electromagnetic charge-response witness under the charge lattice and charge-response ledger, i.e. $q_{\text{EM}}(x) = 0$ as a charge-response witness status. It is a null-load filter on charge response, not primitive standing by the number zero and not a carrier component by itself.

Definition 5.6 (Neutral weak matter carrier).

$$\begin{aligned} C_{\text{neutral-weak-matter}}(x) := & C_{\text{matter-rep}}(x) \wedge \text{ChargeNeutral}(x) \wedge C_{\text{weak-chiral}}(x) \\ & \wedge C_{\text{mass-response-slot}}(x). \end{aligned}$$

Flavor labels, oscillation baselines, PMNS coordinates, and matter-effect routes are not primitive components of this carrier. They occupy a licensed neutrino flavor route/interface status.

Remark 5.7 (Null-load filters are not carriers). ColorNull and ChargeNeutral restrict already-admitted matter-response branches. They do not behave like additional primitive carrier components. The charged-lepton and neutrino carrier formulas use them to state which color or charge load is absent under the fixed ledgers; the primitive standing remains the matter-response role licensed by representation, charge, weak, and mass-response discipline.

Definition 5.8 (Neutrino flavor route/interface status).

$$\begin{aligned} R_{\nu, \text{flavor}} = \{ & \text{weak flavor boundary labels, oscillation baselines, PMNS coordinates,} \\ & \text{mass-splitting routes, matter-effect routes} \}. \end{aligned}$$

Members of $R_{\nu, \text{flavor}}$ may witness, route, summarize, or project neutral weak matter standing. They do not self-promote into primitive neutrino standing.

Definition 5.9 (Mass-response-slot carrier). $C_{\text{mass-response-slot}}(x)$ is not a mass value, Yukawa entry, eigenvalue, pole mass, running mass, constituent mass, or numerical ratio. It is the licensed role by which matter standing admits deformation, threshold, and inertial response through Higgs bridge/fixation and Yukawa/mass compatibility machinery.

Proposition 5.10 (No primitive matter carrier by mass or Yukawa surface).

$$\text{MassValue}(m) \vee \text{YukawaEntry}(y_f) \implies \text{MatterSkin}(\cdot)$$

unless an independent matter carrier certificate is supplied. No m_f , y_f , v , ratio, pole scheme, or low-energy mass parameter selects primitive matter identity by itself.

6 Matter carrier certificates and skin identification

Definition 6.1 (Matter carrier certificate). *A closed matter carrier certificate for an individual matter carrier is*

$$\text{MCert}(x) := \text{RoleCl}(x) \wedge \text{RedInv}(x) \wedge \text{Lic}(x) \wedge \text{NoDesc}(x) \wedge \text{NoSel}(x) \wedge \text{MClose}(x).$$

Here RoleCl closes the load-bearing matter role; RedInv proves invariance under lawful redescription; Lic proves the licensing relation to ordinary skins; NoDesc prevents primitive standing from being assigned to a downstream status; NoSel blocks hidden selector import from a skin; and MClose supplies the local matter-sector closure theorem. For an individual carrier C_i , $\text{ClosedMCert}(C_i)$ means that this certificate has been discharged by the local proof obligations stated in Section 8.

Definition 6.2 (Package-level closed matter certificate). *For the package*

$$C_{\text{matter}} = \{C_{\text{matter-rep}}, C_{\text{color-bearing-matter}}, C_{\text{colorless-charged-matter}}, \\ C_{\text{neutral-weak-matter}}, C_{\text{mass-response-slot}}\},$$

$\text{ClosedMCert}(C_{\text{matter}})$ abbreviates the package-level certificate

$$\text{ClosedMCert}(C_{\text{matter}}) := \bigwedge_{C_i \in C_{\text{matter}}} \text{ClosedMCert}(C_i) \wedge \text{Coverage}_{D_{\text{matter}}^{N_g}}(C_{\text{matter}}) \\ \wedge \text{HostileResidual}_{D_{\text{matter}}^{N_g}}(C_{\text{matter}}).$$

The coverage clause is discharged by the Matter Coverage Theorem rather than by stipulation. It states that the five listed carriers are the complete primitive-role carrier register for same-scope matter claims under the fixed family envelope once the classifier, skin atlas, route atlas, generation branch, and BSM certificates are applied. The hostile-residual clause states that a non-descending primitive assertion outside C_{matter} is exposed as burden or as an explicit hostile-certificate branch; it is not absorbed into a registered status.

Definition 6.3 (Hostile certificate branch). *For a same-scope matter claim x ,*

$$\text{HostileCertificateBranch}(x) := \text{ClosedMCert}(x) \wedge x \not\Downarrow C_{\text{matter}} \wedge \text{NonDerivative}(x).$$

This branch is not a successful registered status inside SM-3. If it closes while preserving $D_{\text{matter}}^{N_g}$, it defeats the scoped matter deletion theorem rather than being reclassified away.

Proposition 6.4 (Certificate sufficiency).

$$\text{MCert}(x) \Rightarrow \text{Carrier}_{\text{matter}}(x).$$

Proof. The certificate contains role closure, redescription invariance, licensing, descent exclusion, no-hidden-selector proof, and local closure. None of these clauses assumes carrier status as a premise. Together they provide the matter-sector analogue of the SM-1 carrier-certificate sufficiency rule. Hence a claim with $\text{MCert}(x)$ occupies carrier standing in the matter sector. $\square$

Definition 6.5 (Matter skin). *A submitted matter item s is eligible for matter-skin classification when its status depends on spinor notation, field label, representation label, algebraic vector-space realization, basis choice, chirality label, mass value, Yukawa coordinate, flavor name, generation label, detector route, hadron membership, parton distribution, oscillation route, PMNS coordinate, cosmological projection, EFT coordinate, BSM model name, or failure of invariant matter-role preservation under admissible redescription.*

Theorem 6.6 (Matter skin non-promotion).

$$\text{MatterSkin}(s) \wedge \neg \text{MCert}(s) \Rightarrow \neg \text{Primitive}_{\text{matter}}(s).$$

Proof. The SM-1 no-promotion theorem states that an object satisfying skin-identification tests cannot become primitive without an independent carrier certificate [2]. The matter-sector skin tests above identify dependence on field notation, representation display, mass coordinate, flavor route, detector route, projection, model surface, or lack of invariant role preservation. Primitive matter standing requires role-stable preservation inside $D_{\text{matter}}^{N_g}$. Without $\text{MCert}(s)$, no independent matter carrier has been supplied to lift the item from derivative status into primitive standing. Therefore s is nonprimitive in the matter arena. $\square$

7 Matter carrier register

Table 2: Closed matter carrier register

Carrier	Primitive role	Licensed skins	Serious falsifier
$C_{\text{matter-rep}}$	Representation-indexed matter-response standing	Spinor fields, representation labels, multiplet notation, chirality notation	Same-scope matter standing outside representation-response carrier discipline.
$C_{\text{color-bearing-matter}}$	Quark standing as color-bearing matter response under color transport	Quark fields, color labels, parton routes, hadron membership, current/constituent mass labels	Free quark with same-scope primitive SM standing.
$C_{\text{colorless-charged-matter}}$	Charged-lepton standing as colorless charge-response matter	Electron/muon/tau labels, pole masses, tracks, lifetimes, decay routes	Charged-lepton primitive outside colorless charged matter-response carrier.
$C_{\text{neutral-weak-matter}}$	Neutrino standing as neutral weak matter response	Neutrino field labels, weak-interaction boundary events, mass-response slots, flavor-route witnesses	Same-scope primitive neutrino outside neutral weak matter carrier.
$C_{\text{mass-response-slot}}$	Licensed mass-response slot over Higgs/Yukawa/mass bookkeeping	Mass values, Yukawa entries, pole/current/constituent masses, ratios	Mass or Yukawa coordinate supplies primitive matter identity without carrier descent.

The derivative neutrino flavor interface is not part of C_{matter} . It is recorded separately as $R_{\nu, \text{flavor}}$, which licenses weak flavor boundary labels, oscillation baselines, PMNS coordinates, mass-splitting routes, and matter-effect routes as route, measurement, mixing, or projection statuses.

8 Local matter carrier certificate discharge

This section makes the proof spine proof-active. Each carrier is checked against the certificate fields

$$\text{RoleCl}, \quad \text{RedInv}, \quad \text{Lic}, \quad \text{NoDesc}, \quad \text{NoSel}, \quad \text{MClose}.$$

8.1 Certificate for $C_{\text{matter-rep}}$

Obligation	Discharge
RoleCl	Representation-response standing is supplied by the pre-linear A4 response ledger, representation-response rigidity, and the fixed matter-response structure.
RedInv	The role is stable under spinor, basis, algebraic representation, chiral, tensor, dimension, and multiplicity redescription.
Lic	The carrier licenses spinor fields, multiplet labels, representation labels, chiral labels, and algebraic representation spaces.
NoDesc	The role is not reducible to any one spinor notation, basis, multiplet label, dimension count, or algebraic representation.
NoSel	No preferred basis, representation label, dimension, chirality label, or algebraic realization selects matter identity.
MClose	Closure uses A4, PSM3, SR3, RPI, and the SM-1 classifier.

The representation carrier closes because the pre-linear response ledger supplies the matter-response role, while PSM3 licenses algebraic realization only as witness-preserving representative language [6, 7, 8]. Thus spinor and representation notation are licensed without becoming primitive.

8.2 Certificate for $C_{\text{color-bearing-matter}}$

Obligation	Discharge
RoleCl	Color-bearing matter response is the intersection of matter-representation response with the SM-2 color-transport carrier under confinement-compatible discipline.
RedInv	The role is stable under color labels, quark field notation, parton routes, hadron membership, jets, and scattering displays.
Lic	It licenses quark spinors, color labels, current/constituent masses, parton distributions, hadron membership, jets, and scattering route language.
NoDesc	Quark primitive standing is not reducible to isolated-particle skin, jet, hadron membership, color index, or parton route.

NoSel	No detector jet, parton route, color index, or hadron membership selects primitive quark identity.
MClose	Closure uses $C_{\text{matter-rep}}$, SM-2 color transport, and ConfCompat.

SM-2 closes color transport; SM-3 adds the matter-side representation-response carrier. Quark standing closes only at their intersection, under confinement-compatible discipline.

8.3 Certificate for $C_{\text{colorless-charged-matter}}$

Obligation	Discharge
RoleCl	Colorless charged matter-response standing is representation response plus color-null status, charge response, weak-chiral response, and mass-response-slot standing.
RedInv	The role is stable under spinor notation, track routes, pole masses, decay channels, lifetimes, and Yukawa/mass bookkeeping.
Lic	It licenses electron/muon/tau labels, charged-lepton fields, pole masses, tracks, decay routes, lifetimes, and mass-shape readouts.
NoDesc	Charged-lepton primitive standing is not any one mass, track, lifetime, decay route, or label.
NoSel	No pole mass, track, family index, or numerical readout selects primitive charged-lepton identity.
MClose	Closure uses representation response, charge lattice, SM-2 charge/weak support, and mass-response-slot support.

The charged-lepton support papers are used only to discipline mass-shape, role-projection readout, one-anchor import, and readout-boundary statuses. They do not ground the charged-lepton primitive carrier and do not convert SM-3 into charged-lepton numerical closure [15, 16, 17].

8.4 Certificate for $C_{\text{neutral-weak-matter}}$

Obligation	Discharge
RoleCl	Neutral weak matter-response standing is representation response plus charge-neutral status, weak-chiral response, and mass-response-slot standing.
RedInv	The role is stable under neutrino field labels, weak flavor boundary labels, mass eigenstate notation, oscillation routes, PMNS coordinates, and cosmological projections.
Lic	It licenses neutrino fields, weak flavor labels, oscillation baselines, mass splittings, PMNS entries, missing-energy routes, and cosmological bounds.
NoDesc	Neutrino primitive standing is not a weak flavor label, oscillation route, PMNS coordinate, mass splitting, or cosmological bound.

NoSel	No PMNS convention, baseline, missing-energy route, cosmological bound, or sterile label selects primitive neutrino identity.
MClose	Closure uses representation response, SM-2 weak-chiral standing, charge-neutral discipline, mass-response-slot standing, and neutrino reconstruction route discipline.

Flavor is not part of the primitive neutrino carrier. It is a licensed boundary/route interface attached to neutral weak matter standing.

8.5 Certificate for $C_{\text{mass-response-slot}}$

Obligation	Discharge
RoleCl	The slot is a bridge-licensed deformation, threshold, and inertial-response role for matter under Higgs/Yukawa/mass compatibility.
RedInv	The role is stable across Higgs-Yukawa and low-energy mass bookkeeping.
Lic	It licenses pole masses, running masses, current masses, constituent masses, Yukawa entries, mass ratios, and threshold displays.
NoDesc	The slot is not reducible to any one numerical mass value, pole scheme, low-energy mass value, or Yukawa coordinate.
NoSel	No m_f , y_f , v , ratio, pole scheme, or low-energy mass parameter selects primitive matter identity.
MClose	Closure uses Higgs bridge/fixation, Yukawa compatibility, L1 Yukawa/mass equivalence, and matter-representation standing.

L1 Paper 3 supplies the negative overcount collapse; Yukawa compatibility and Higgs bridge support supply the positive slot-standing role [12, 13]. Neither supplies numerical mass eigenvalue closure.

9 Representation-response matter standing

9.1 Representation response as the root matter role

Matter standing first enters through representation response, not through a spinor field symbol. The A4 response ledger records representation response and charge-witness standing before algebraic representation spaces are admitted [6]. The matter-representation carrier is therefore

$$C_{\text{matter-rep}} = \text{RepResp}_{\text{matter}}.$$

9.2 Algebraic representation bridge

Representation algebra specialization supplies the bridge

$$\text{Resp}_D^\partial + \text{AlgRef}_D^\partial \Rightarrow \text{RepAlgSpec}_D^\partial \subseteq A_D^\partial$$

under the already licensed transition, transport, loop-residue, and product-sector inputs [7]. This result licenses vector spaces, modules, group actions, Lie-algebra actions, direct sums, tensor products,

duals, conjugates, chiral splits, dimensions, and multiplicities only as response-witness-preserving algebraic realizations.

Thus algebraic representation language is available, but it is not primitive internal matter structure. A spinor space, multiplet notation, chiral decomposition, tensor product, or dimension label carries matter status only insofar as it realizes the response ledger without basis privilege, hidden multiplicity, or scope expansion.

Theorem 9.1 (No primitive spinor theorem).

$$\text{SpinorField}(\psi) \wedge \neg \text{MCert}(\psi) \Rightarrow \neg \text{Primitive}_{\text{matter}}(\psi).$$

Proof. A spinor field is algebraic and representational notation. By representation algebra specialization, algebraic representation language is admissible only as a witness-preserving realization of the response ledger [7]. If ψ lacks a matter carrier certificate, it has not supplied a role-stable primitive matter carrier. Therefore ψ remains a licensed representative or fails primitive promotion. $\square$

Theorem 9.2 (No primitive representation-label theorem).

$$\text{RepLabel}(r) \wedge \neg \text{MCert}(r) \Rightarrow \neg \text{Primitive}_{\text{matter}}(r).$$

Labels such as doublet, singlet, triplet, left, right, fundamental, conjugate, or generation become standing-bearing only when licensed by the response ledger.

Theorem 9.3 (Chiral label status). *A chiral label has matter standing only when it is a witnessed handed response distinction. If it is merely a formal component tag or basis-dependent split, it is a matter skin.*

Proof. Representation algebra specialization admits chiral response sectors only when response witnesses distinguish handed response in a quotient-stable, transport-compatible, loop-compatible manner [7]. A chiral label lacking such witness structure cannot be primitive. It is either a representative, a downstream route, hidden load, or a failed primitive promotion. $\square$

10 Field, particle, and persistence interface

The realized physical interior supplies the domain in which fields and particles are not free vocabulary. Fields are persistence carriers, and particles are localized persistence classes inside a boundary-licensed physical stack [5]. SM-3 uses this to keep matter claims physical while denying primitive standing by notation alone.

Proposition 10.1 (Fields as matter representatives).

$$\text{FieldSymbol}(\psi) \not\Rightarrow \text{Primitive}_{\text{matter}}(\psi).$$

A matter field is lawful when it represents matter-response standing. It is not primitive by field notation alone.

Proposition 10.2 (Particles as localized persistence classes).

$$\text{ParticleLabel}(p) \not\Rightarrow \text{Primitive}_{\text{matter}}(p).$$

Particle labels are lawful when they name localized persistence classes licensed by matter carriers. The name does not settle primitive matter identity.

Criterion 10.3 (Matter identity criterion). *A matter claim has primitive standing only when it preserves representation response, the relevant gauge/color/charge/weak role, field-carrier mediation, particle-persistence identity, boundary witness, transport/redescription invariance, and no hidden selector from a skin.*

11 Quark standing

11.1 Submitted quark claims

Quark claims arrive through quark field notation, color labels, flavor labels, current masses, constituent masses, hadron membership, parton distributions, scattering routes, jets, confinement routes, and isolated free-quark assertions. The classifier treats each as a submitted claim rather than as an automatic primitive.

11.2 Why color transport alone does not close quark matter standing

SM-2 closes color transport and gluon non-asymptotic standing. That does not by itself close quark matter standing. Quark standing requires the intersection of matter representation-response with color transport under confinement-compatible discipline. Quark labels, parton distributions, jets, hadron membership, current masses, and constituent masses are licensed displays or routes. A free-quark claim is the hostile branch, not the default primitive.

11.3 Quark carrier

$$C_q = C_{\text{color-bearing-matter}} = C_{\text{matter-rep}} \wedge C_{\text{color-transport}} \wedge \text{ConfCompat}.$$

Quark standing is color-bearing matter standing under confinement-compatible target discipline. The color side is received from SM-2; the matter side is closed here.

11.4 Quark skin atlas

Quark-associated item	SM-3 status
Quark spinor field	Licensed representative of color-bearing matter standing.
Color label	Representation/color-transport skin.
Flavor name	Route or representation label; SM-5 boundary when transition standing is implicated.
Current mass	Mass-response and renormalization status.
Constituent mass	Effective/projection route status.
Hadron membership	Confinement route status.
Parton distribution	Projection/route status.
Jet association	Detector/projection route.
Isolated-particle picture	False primitive unless a same-scope carrier certificate closes.

Theorem 11.1 (Quark color-bearing matter carrier theorem). *If $\text{QuarkClaim}(x) \wedge \text{SameScope}_{D_{\text{matter}}^{N_g}}(x)$, then*

$$x \downarrow C_{\text{color-bearing-matter}} \vee \text{Rep}(x) \vee \text{Meas}(x) \vee \text{Route}(x) \vee \text{Proj}(x) \vee \text{Trans}(x) \vee \text{Burden}(x) \vee \text{FailPrim}(x).$$

Proof. Same-scope quark claims preserve $D_{\text{matter}}^{N_g}$. The representation-response root supplies the matter side, and SM-2 supplies the color-transport role while blocking primitive isolated color skins [3, 6, 7, 8]. Confinement-compatible discipline does not prove quark closure by stipulation; it records that a candidate isolated-quark primitive must defeat both color-transport descent and matter-response descent under the same fixed target. Hadron membership, parton routes, jets,

and scattering displays are therefore route or projection statuses unless a hostile free-color matter certificate closes. This discipline does not prove quark closure by definition; it fixes the branch structure. The closure proof uses $C_{\text{matter-rep}}$ for the matter side and SM-2 $C_{\text{color-transport}}$ for the color side. A free isolated-quark claim would have to defeat both descents while preserving the same target. Current and constituent masses satisfy mass-response or effective-projection dependence. If a claim supplies a closed carrier certificate, it descends to $C_{\text{color-bearing-matter}}$. If not, the classifier assigns representative, measurement, route, projection, transport, burden, or failed primitive status. $\square$

Corollary 11.2 (Quark primitive-status deletion).

$$\text{Survprim}_{D_q}(\text{Attr}_q \setminus C_{\text{color-bearing-matter}}) = 0.$$

Criterion 11.3 (Free-quark falsifier). *A free-quark falsifier must satisfy*

$$\text{FreeQuark}(y) \wedge \text{SameScope}_{D_{\text{matter}}^{N_g}}(y) \wedge \text{MCert}(y) \wedge y \not\downarrow C_{\text{color-bearing-matter}}.$$

A detector anomaly, jet, parton route, confinement stress, or unusual hadronization pattern is not sufficient by itself.

12 Charged-lepton standing

12.1 Submitted charged-lepton claims

Charged-lepton claims arrive through electron, muon, and tau labels; spinor fields; electric charge; pole masses; decay routes; lifetimes; tracks; mass ratios; Yukawa entries; family labels; and numerical readout claims.

12.2 What charged-lepton support is used for

The charged-lepton support is used to discipline mass-shape, role-projection readout, one-anchor import, and readout-boundary status. It is not used as the primitive ground of charged-lepton standing and is not invoked as exact pole-mass closure. The primitive carrier is the colorless charged matter-response carrier; mass-shape and readout objects remain licensed statuses.

12.3 Charged-lepton carrier

$$C_\ell = C_{\text{colorless-charged-matter}}$$

where

$$\begin{aligned} C_{\text{colorless-charged-matter}} &= C_{\text{matter-rep}} \wedge \text{ColorNull} \wedge C_{\text{charge-response}} \\ &\quad \wedge C_{\text{weak-chiral}} \wedge C_{\text{mass-response-slot}}. \end{aligned}$$

Charged leptons are colorless charged matter-response carriers with weak-chiral and mass-response licensing. Here $C_{\text{weakext-chiral}}$ denotes the admissible chiral-load envelope for the charged-lepton matter role, including the left/right representation-response split and Yukawa/mass compatibility. It does not mean that every charged-lepton representative carries identical weak response.

12.4 Charged-lepton skin atlas

Charged-lepton item	SM-3 status
Electron, muon, tau labels	Licensed matter labels.
Spinor fields	Representatives of matter-response standing.
Electric charge value	Charge-response witness or calibration.
Pole mass	Measurement witness and mass-response skin.
Lifetime	Route/measurement status.
Decay channel	Route status.
Detector track	Measurement/projection witness.
Yukawa entry	Mass-response bookkeeping or coupling slot.
Mass ratio	Downstream readout or boundary status.
Family index	Fixed-boundary or role-index label, not primitive identity.

Theorem 12.1 (Charged-lepton colorless charged matter theorem). *If $\text{ChargedLeptonClaim}(x) \wedge \text{SameScope}_{D_{\text{matter}}^{N_g}}(x)$, then*

$$x \downarrow C_{\text{colorless-charged-matter}} \vee \text{Rep}(x) \vee \text{Meas}(x) \vee \text{Route}(x) \vee \text{Proj}(x) \\ \vee \text{Trans}(x) \vee \text{Burden}(x) \vee \text{FailPrim}(x).$$

Proof. The matter-representation role is supplied by representation response and algebraic specialization. ColorNull fixes absence of color-transport load without adding a primitive substance. Charge-response and weak-chiral roles are received from SM-2, L1 charge support, and anomaly/Yukawa compatibility [3, 10, 11]. Mass and Yukawa surfaces descend to the mass-response slot by the Yukawa/mass equivalence and Yukawa-compatibility support [12, 13]. Tracks, lifetimes, and decays are measurement or route statuses. Therefore the only carrier branch is descent to $C_{\text{colorless-charged-matter}}$; all other same-scope charged-lepton surfaces receive registered derivative, burden, or failure status. $\square$

Theorem 12.2 (Pole mass non-primitivity).

$$\text{PoleMass}(m_\ell) \wedge \neg \text{MCert}(m_\ell) \Rightarrow \neg \text{Primitive}_\ell(m_\ell).$$

Proof. A pole mass is a measurement witness and mass-response skin. It can witness charged-lepton matter response, but it does not occupy the colorless charged matter carrier role. Without an independent carrier certificate, it cannot ground primitive charged-lepton identity. $\square$

Nonclaim 12.3 (Charged-lepton numerical boundary). *SM-3 uses the charged-lepton arc only for charged-lepton residue, role-projection, readout, and mass-shape status discipline [15, 16, 17]. It does not assert exact charged-lepton pole-value derivation as part of the matter-carrier theorem.*

Criterion 12.4 (Charged-lepton falsifier). *A charged-lepton falsifier must supply same-scope primitive lepton standing outside $C_{\text{colorless-charged-matter}}$. A new mass fit, track anomaly, decay anomaly, or Yukawa coordinate does not suffice unless it closes a matter carrier certificate outside the carrier package.*

13 Neutrino standing

13.1 Submitted neutrino claims

Neutrino claims arrive through neutrino field labels, flavor labels, mass eigenstate labels, oscillation baselines, PMNS entries, mass splittings, absolute mass bounds, matter effects, ordering claims, sterile-neutrino labels, cosmological constraints, and weak-interaction boundary events.

13.2 Neutrino carrier

$$C_\nu = C_{\text{neutral-weak-matter}}$$

where

$$C_{\text{neutral-weak-matter}} = C_{\text{matter-rep}} \wedge \text{ChargeNeutral} \wedge C_{\text{weak-chiral}} \\ \wedge C_{\text{mass-response-slot}}.$$

Neutrino standing is neutral weak matter standing. Weak flavor, oscillation, PMNS, mass-splitting, and cosmological surfaces are licensed route, mixing, measurement, or projection statuses attached to that carrier. SM-3 does not assert that the minimal textbook massless-neutrino presentation already contains numerical neutrino mass closure. The carrier claim is that any same-scope neutrino matter standing involving mass response is typed through the licensed mass-response slot; absolute mass values, ordering, Dirac/Majorana status, and PMNS parameters remain downstream or burdened.

13.3 Flavor as route and boundary status

The neutrino reconstruction support treats flavor under noncommuting propagation as a boundary-kernel reconstruction rather than a globally conserved primitive flavor identity [18]. SM-3 uses this result to preserve oscillation physics while preventing flavor labels from becoming primitive matter identity.

$$\text{FlavorLabel}(\alpha) \not\Rightarrow \text{Primitive}_\nu(\alpha).$$

Oscillation is a route/reconstruction status, not primitive flavor identity by route availability alone. More formally,

$$\text{FlavorRoute}_\nu(r) \wedge \neg \text{MCert}(r) \Rightarrow \neg \text{Primitive}_\nu(r).$$

13.4 Neutrino skin atlas

Neutrino item	SM-3 status
Neutrino field label	Licensed representative.
Weak flavor label	Boundary/route label.
Mass eigenstate label	Mass-response/route status.
PMNS entry	Mixing-coordinate skin; SM-5 boundary.
Oscillation baseline	Route/measurement status.
Mass splitting	Measurement/route witness.
Absolute mass bound	Cosmological projection status.
Ordering claim	Observable-transport burden or prediction-lock status.
Sterile-neutrino label	Route, extension, burden, target update, or failed primitive.
Missing energy	Detector/projection witness.

Theorem 13.1 (Neutrino neutral weak matter theorem). *If $\text{NeutrinoClaim}(x) \wedge \text{SameScope}_{D_{\text{matter}}^{N_g}}(x)$, then*

$$x \downarrow C_{\text{neutral-weak-matter}} \vee \text{Rep}(x) \vee \text{Meas}(x) \vee \text{Route}(x) \vee \text{Mix}(x) \\ \vee \text{Proj}(x) \vee \text{Trans}(x) \vee \text{Burden}(x) \vee \text{FailPrim}(x).$$

Proof. Neutrino field notation descends to representation-response standing; weak response descends to the SM-2 weak-chiral carrier; charge-neutral status descends to charge-response and charge-lattice discipline; mass surfaces descend to the mass-response slot; oscillation surfaces descend to boundary-to-boundary reconstruction and route status; PMNS entries and phase conventions are mixing-coordinate statuses reserved for SM-5; cosmological constraints are projection statuses. A sterile label is not a carrier certificate. Hence every same-scope neutrino claim either descends to $C_{\text{neutral-weak-matter}}$ or receives registered derivative, burden, or failure status. $\square$

Nonclaim 13.2 (Neutrino numerical boundary). *SM-3 does not prove the absolute neutrino mass scale, normal/inverted ordering closure, PMNS numerical values, Dirac/Majorana status closure, or sterile-neutrino model exclusion. Neutrino ordering and matter-envelope support are used as observable-transport and route/projection discipline, not as full flavor-sector closure [22, 23, 24, 25, 26, 27, 28].*

Criterion 13.3 (Sterile-neutrino branch). *A sterile-neutrino claim is classified as*

$$\text{Route} \vee \text{Ext} \vee \text{Mix} \vee \text{Proj} \vee \text{Burden} \vee \text{TargetUpdate} \vee \text{FailPrim}$$

unless it supplies

$$\text{SameScope}_{D_{\text{matter}}^{N_g}}(s) \wedge \text{MCert}(s) \wedge s \not\downarrow C_{\text{neutral-weak-matter}} \wedge \text{NonDerivative}(s).$$

14 Mass-response and Yukawa boundary

14.1 Mass is not matter identity

A mass value can witness matter response, fix a threshold, enter a propagator, participate in a pole scheme, or serve as a low-energy parameter. It is not primitive matter identity by numerical value alone:

$$\text{MassLabel}(m) \not\Rightarrow \text{Primitive}_{\text{matter}}(m).$$

14.2 Negative theorem: no primitive identity by mass or Yukawa coordinate

Theorem 14.1 (Mass/Yukawa coordinate non-promotion).

$$(\text{MassValue}(m) \vee \text{YukawaEntry}(y_f)) \wedge \neg \text{MCert}(\cdot) \Rightarrow \neg \text{Primitive}_{\text{matter}}(\cdot).$$

Proof. SM-1 blocks primitive standing from values and parameters without a carrier certificate. L1 Yukawa/mass equivalence shows that Higgs-Yukawa and low-energy massive-fermion descriptions carry a single invariant witness per fermion species under admissible transport [12]. Treating (y_f, v) and m_f as independent primitive matter-standing carriers would overcount witness load. Thus mass and Yukawa coordinates are skins, measurement witnesses, transport coordinates, or mass-response-slot statuses unless independently certified. $\square$

14.3 Positive theorem: mass-response-slot standing

The L1 Yukawa/mass equivalence paper supplies the eliminative relation between Higgs-Yukawa bookkeeping and low-energy massive-fermion bookkeeping [12]. In standard notation,

$$m_f \leftrightarrow \frac{y_f v}{\sqrt{2}}.$$

The SM-3 use of this relation is not numerical prediction. It is invariant witness counting: (y_f, v) does not supply two independent primitive matter carriers when the low-energy fermion sector carries one invariant mass witness.

Theorem 14.2 (Mass-response slot theorem). *If $MassOrYukawaClaim(x) \wedge \text{SameScope}_{D_{\text{matter}}^{N_g}}(x)$, then*

$$x \downarrow C_{\text{mass-response-slot}} \vee \text{Trans}(x) \vee \text{Meas}(x) \vee \text{Route}(x) \vee \text{Burden}(x) \vee \text{FailPrim}(x).$$

Proof. The positive slot-standing role is licensed by Higgs bridge/fixation, Yukawa compatibility, representation-response matter standing, and admissible transport between Higgs-Yukawa and low-energy mass bookkeeping [13, 12]. The slot is a carrier role for mass response; it is not a numerical mass value. Therefore mass and Yukawa claims either descend to the mass-response slot or occupy transport, measurement, route, burden, or failed primitive status. $\square$

Corollary 14.3 (No primitive Yukawa-entry theorem).

$$YukawaEntry(y_f) \wedge \neg \text{MCert}(y_f) \Rightarrow \neg \text{Primitive}_{\text{matter}}(y_f).$$

Nonclaim 14.4 (Mass-response boundary). *The mass-response slot theorem does not derive m_f , y_f , v , pole masses, current masses, constituent masses, mass ratios, or exact lepton or neutrino mass values. It closes only the primitive-status role of the mass-response slot.*

14.4 Boundary to SM-4 and SM-5

Coupling/RG behavior belongs to SM-4. Flavor/mixing coordinates belong to SM-5. SM-3 closes only the primitive matter-identity promotion route for mass and Yukawa surfaces.

15 Generation and family-label discipline

15.1 Generation labels

Generation labels do not carry primitive matter identity by label alone:

$$GenerationLabel(g) \wedge \neg \text{MCert}(g) \Rightarrow \neg \text{Primitive}_{\text{matter}}(g).$$

15.2 Fixed-boundary status

Generation-count support is used as fixed-boundary classification discipline [14]. The family envelope N_g records the representation regime boundary and blocks arbitrary family duplication as same-scope primitive matter standing. SM-3 does not present generation number as a new numerical prediction.

15.3 Generation-cardinality altering claims

Let $GenCardAlter(z)$ mean that a submitted claim changes the family-cardinality envelope rather than merely using a generation label inside the fixed arena. Then

$$GenCardAlter(z) \Rightarrow Burden(z) \vee FailPrim(z) \vee TargetUpdate(z)$$

unless a role-cardinality certificate closes. A field copy, mass value, new mixing coordinate, or family label is not such a certificate. Full role-cardinality closure is an SM-5/SM-6 burden unless a dedicated module supplies the certificate.

16 Detector, route, and projection atlas

Detector and route evidence is preserved but typed. Measurements witness carriers or routes; they do not generate primitive matter standing.

Table 8: Matter detector, route, and projection atlas

Object	Status	Non-promotion rule
Detector track	Measurement/projection witness	A track does not create matter identity.
Jet	Route/projection witness	A jet does not create primitive quark or gluon identity.
Missing energy	Detector/projection route	Missing energy does not create primitive neutrino standing.
Hadron membership	Confinement route	Membership does not ground primitive quark identity.
Oscillation baseline	Route/measurement	A baseline does not ground primitive neutrino identity.
Cosmological bound	mass Projection status	A cosmological bound does not ground primitive neutrino identity.
Decay topology	Route status	A decay route does not create primitive matter standing.
Scattering signature	Route/measurement	Scattering success does not self-ground a carrier.
EFT fermion operator	Projection/transport status	An operator coefficient does not create primitive matter identity.

17 BSM and EFT branch certificates

17.1 Branch logic

For any matter-sector BSM claim b , the same-scope branch is theorem-clean only when the hostile carrier branch is displayed explicitly:

$$\begin{aligned} & \text{MatterBSMClaim}(b) \wedge \text{SameScope}_{D_{\text{matter}}^{N_g}}(b) \Rightarrow \\ & \left[\text{RegisteredStatus}(b) \vee \text{Burden}(b) \vee \text{FailPrim}(b) \right. \\ & \quad \vee (\text{ClosedMCert}(b) \wedge b \not\Downarrow C_{\text{matter}} \\ & \quad \left. \wedge \text{NonDerivative}(b)) \right]. \end{aligned}$$

The final disjunct is the local falsifier branch. If the claim is not same-scope, then

$$\text{MatterBSMClaim}(b) \wedge \neg \text{SameScope}_{D_{\text{matter}}^{N_g}}(b) \Rightarrow \text{TargetUpdate}(b).$$

Here RegisteredStatus abbreviates representative, measurement, transport, route, mixing, projection, and extension dispositions.

17.2 Full sterile-neutrino branch tree

A sterile-neutrino claim is the most direct stress test for the neutrino carrier. The classifier applies the following branch sequence.

Table 9: Sterile-neutrino branch certificate

Stage	Test	Terminal branch
Target test	Does the claim preserve $D_{\text{matter}}^{N_g}$, weak-chiral response, charge-neutral status, mass-response slot, source state, and no-hidden-sector load?	If not, TargetUpdate.
Carrier test	Does it supply neutral weak matter standing outside C_ν ?	If certified, hostile carrier branch.
Descent test	Does it descend to oscillation route, weak-flavor boundary label, mass-state skin, PMNS/mixing coordinate, cosmological projection, or detector missing-energy route?	Route, Mix, Proj, or Meas.
Extension test	Is it a carrier-preserving extension disposition under fixed matter carrier roles?	Ext.
Burden test	Is same-scope primitive standing asserted but not certified?	Burden.
Failure test	Is primitive status asserted only from the label, mass anomaly, oscillation anomaly, or cosmological bound?	FailPrim.
Falsifier branch	Does it preserve same scope, defeat derivative descent, and close MCert outside C_ν ?	Local SM-3 falsifier.

The falsifier condition is deliberately strict:

$$\text{SameScope}_{D_{\text{matter}}^{N_g}}(s) \wedge \text{ClosedMCert}(s) \wedge \text{NonDerivative}(s) \wedge s \not\Downarrow C_\nu.$$

A sterile label, oscillation anomaly, mass-state surface, or cosmological bound is not sufficient by itself.

17.3 Matter BSM branch matrix

Table 10: Matter BSM/EFT branch matrix

Claim type	Target/descent tests	Expected branch	Falsifier
Sterile neutrino	Weak-flavor link, mass-state skin, cosmological projection, hidden-sector load	Route, extension, burden, target update, or failed primitive	Same-scope primitive neutrino carrier outside C_ν .
Vector-like fermion	Chiral-load test, anomaly ledger, charge lattice, mass-response slot, target update boundary	Extension, burden, target update, or failed primitive	Same-scope primitive matter carrier preserving chiral load, anomaly, charge, and mass-slot ledgers.
Fourth generation	Fixed family envelope, anomaly, charge, vacuum, Yukawa realizability, mixing boundary	Burden, failed primitive, or target update	Same-scope fourth generation preserving all ledgers with role-cardinality certificate closed.
Leptoquark	Simultaneous quark/lepton role preservation, color/charge ledger, representation ledger	Extension, projection, route, burden, target update, or failed primitive	Same-scope primitive carrier preserving both quark and lepton roles without hidden ledger change.
Dark-sector fermion	Hidden-source state, carrier inventory, projection interface, charge neutrality	Target update, extension, projection, burden, or failed primitive	Hidden-sector fermion preserving $D_{\text{matter}}^{N_g}$ and closing independent carrier certificate.

Claim type	Target/descent tests	Expected branch	Falsifier
Fermion compositeness	Reconstruction of same carrier or changed substructure target	Projection, route, burden, target update, or failed primitive	Composite reconstruction displaces or supplements C_{matter} under fixed target with certificate closed.
EFT fermion operator	Operator basis, matching, measurement surface, projection pressure	Projection, transport, route, burden, or target update	Operator coefficient supplies primitive matter standing after projection, route, measurement, and transport statuses are defeated.

Theorem 17.1 (Matter BSM branch-certificate theorem). *For any same-scope matter-sector BSM claim b ,*

$$\begin{aligned}
\text{MatterBSMClaim}(b) \wedge \text{SameScope}_{D_{\text{matter}}^{N_g}}(b) \Rightarrow & \text{RegisteredStatus}(b) \vee \text{Burden}(b) \vee \text{FailPrim}(b) \\
& \vee (\text{ClosedMCert}(b) \wedge b \not\downarrow C_{\text{matter}} \\
& \wedge \text{NonDerivative}(b)).
\end{aligned}$$

If b is not same-scope, then $\text{TargetUpdate}(b)$. The final disjunct is the explicit local falsifier branch, not a hidden exception.

Proof. The proof is the SM-1 BSM audit closure applied to $D_{\text{matter}}^{N_g}$, with the local matter carrier package supplied by Sections 9–14. A BSM label, field, symmetry, anomaly, coefficient, or resonance is not a carrier certificate. If target conditions change, target-update status applies. If same scope is preserved but carrier closure does not close, the claim receives a registered status, burden, or failure. Only a closed certificate outside C_{matter} , with derivative descent defeated, occupies the displayed falsifier branch. $\square$

18 Matter prediction box

18.1 Target-preserving prediction and falsification

The prediction register in SM-3 has reality contact only because the target and comparison class are preserved. It is not an internal bookkeeping list. A predicted measurement, anomaly, route, or extension pressure is assessed as a prediction of the same matter-sector claim only when the reference-bearing target in $D_{\text{matter}}^{N_g}$ remains fixed.

Proposition 18.1 (Reality-contact discipline for matter predictions). *For every matter prediction or falsification entry,*

$$\begin{aligned}
& \text{Prediction}_{\text{matter}}(x) \vee \text{Falsification}_{\text{matter}}(x) \\
\Rightarrow & \text{KernelPreserved}_{\text{matter}}(x) \wedge \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x).
\end{aligned}$$

If the target changes, the result is a target update. If the target is preserved but the alleged primitive status does not close, the result is a registered status, burden, or failed primitive promotion rather than a same-scope primitive survivor.

18.2 Sector prediction statement

SM-3's sector prediction is a primitive-status prediction:

$$FutureMatterClaim(y) \Rightarrow \exists \rho_{\text{matter}}(y) \in Status_{\text{matter}} \quad \text{or} \quad Failure(M0).$$

The prediction is falsifiable. A real future matter claim that cannot receive a registered status without grammar expansion defeats M0.

Table 11: Matter prediction register

ID	Claim type	Expected disposition	Falsifier
M0	Any future matter claim	Carrier, representative, measurement, transport, route, mixing, projection, extension, update, burden, or failure	Real matter claim cannot be classified without grammar expansion.
M1	Spinor or field claim	Representative or failure if promoted	Spinor notation alone supplies primitive matter standing.
M2	Quark or free-quark claim	Color-bearing matter standing, route, projection, burden, update, or failure	Free quark with same-scope primitive SM standing.
M3	Charged-lepton claim	Colorless charged matter response, measurement, route, projection, burden, or failure	Charged-lepton primitive outside C_ℓ .
M4	Neutrino claim	Neutral weak matter carrier, plus flavor/oscillation route or mixing/projection status, burden, update, or failure	Neutrino primitive outside C_ν .
M5	Mass or Yukawa claim	Mass-response slot, measurement, transport, route, or failure	Mass/Yukawa coordinate supplies primitive matter identity.
M6	Detector or track claim	Measurement or projection witness	Detector track creates primitive matter standing.
M7	Hadron or parton claim	Confinement route/projection	Hadron membership creates primitive quark identity.
M8	Oscillation or cosmology claim	Route, measurement, projection, or burden	Oscillation or cosmological bound creates primitive neutrino standing.

ID	Claim type	Expected disposition	Falsifier
M9	Generation label or cardinality alteration	Generation labels fail primitive projection; generation-cardinality alterations are burden, failure, or target update unless role-cardinality certificate closes	Same-scope generation addition preserves all matter ledgers with certificate closed.
M10	New fermion label	Representative, extension, projection, burden, update, or failure	New label alone supplies primitive matter carrier.

Theorem 18.2 (Matter prediction box theorem). *No future matter claim receives primitive standing from spinor notation, mass, flavor label, detector track, oscillation route, cosmological bound, current mass, constituent mass, hadron membership, Yukawa entry, generation label, or isolated-particle skin alone.*

Proof. Each listed item satisfies matter-skin criteria. By Theorem 6.6, without MCert it cannot be primitive. A future claim may be empirical, useful, surprising, or severe; it still must pass target, carrier, descent, prediction, burden, and failure tests. $\square$

19 Final scoped matter-sector closure

Theorem 19.1 (Matter-sector classification). *If $\text{MatterClaim}(x) \wedge \text{SameScope}_{D_{\text{matter}}^{N_g}}(x)$, then*

$$\rho_{\text{matter}}(x) \in \text{Status}_{\text{matter}} \setminus \{\text{TargetUpdate}\}.$$

If x does not preserve same scope, target-update status applies.

Theorem 19.2 (No primitive matter-skin escape).

$$\text{MatterSkin}(s) \wedge \neg \text{MCert}(s) \Rightarrow \neg \text{Primitive}_{\text{matter}}(s).$$

In particular, spinor notation, field labels, representation labels, chiral labels, mass values, Yukawa coordinates, flavor names, generation labels, detector tracks, hadron membership, parton routes, oscillation baselines, PMNS entries, cosmological bounds, EFT coefficients, and BSM model names do not self-promote into primitive matter standing.

Theorem 19.3 (Matter Coverage Theorem). *For every same-scope matter claim under the fixed family envelope,*

$$\begin{aligned} \text{MatterClaim}(x) \wedge \text{SameScope}_{D_{\text{matter}}^{N_g}}(x) \Rightarrow x \downarrow C_{\text{matter}} \vee \text{RegisteredStatus}(x) \vee \text{Burden}(x) \\ \vee \text{FailPrim}(x) \vee \text{HostileCertificateBranch}(x). \end{aligned}$$

Here $\text{RegisteredStatus}(x)$ abbreviates licensed representative, measurement witness, transport ledger, route status, mixing status, projection display, or extension disposition.

Proof. The target-update branch is excluded by the same-scope hypothesis. The SM-1 classifier then applies the carrier, descent, prediction, burden, and failure tests. If the carrier test closes through one of the five matter carriers, the claim descends to C_{matter} . If the submitted object is spinor notation, representation notation, mass or Yukawa bookkeeping, a flavor route, detector route,

hadron or parton route, oscillation route, projection surface, or same-scope extension, it receives the appropriate registered status by the matter skin and route atlases. If primitive standing is asserted but the local certificate remains open, the claim is burdened. If primitive standing is asserted from a skin alone, it fails primitive promotion. The only remaining same-scope primitive branch is a closed non-derivative certificate outside C_{matter} , which is exactly $\text{HostileCertificateBranch}(x)$. Thus coverage is a theorem of the fixed classifier applied to the local matter carrier register, not a hidden clause inside the definition of package closure. $\square$

Theorem 19.4 (Matter-representation certificate closure).

$$\text{ClosedMCCert}(C_{\text{matter-rep}}).$$

Proof. The certificate for $C_{\text{matter-rep}}$ is discharged in Section 8. A4 supplies pre-linear representation response; PSM3 licenses algebraic representation language only as witness-preserving realization; SR3 supplies response rigidity; RPI supplies the physical field/particle substrate; SM-1 supplies the classifier. The result is a closed carrier certificate for representation-indexed matter-response standing. $\square$

Theorem 19.5 (Color-bearing matter certificate closure).

$$\text{ClosedMCCert}(C_{\text{color-bearing-matter}}).$$

Proof. The certificate for $C_{\text{color-bearing-matter}}$ is discharged in Section 8. It combines the matter-representation carrier with SM-2 color transport; ConfCompat supplies the branch discipline rather than an independent confinement proof. Quark skins such as color labels, parton routes, hadron membership, jets, and isolated-particle pictures are either licensed statuses or hostile burden branches. A free-color primitive would have to defeat both matter-response descent and color-transport descent under fixed $D_{\text{matter}}^{N_g}$. No primitive quark carrier remains outside the color-bearing matter role under that test. $\square$

Theorem 19.6 (Colorless charged matter certificate closure).

$$\text{ClosedMCCert}(C_{\text{colorless-charged-matter}}).$$

Proof. The certificate for $C_{\text{colorless-charged-matter}}$ is discharged in Section 8. It combines matter-representation standing, ColorNull, charge response, weak-chiral response, and the mass-response slot. Pole masses, tracks, lifetimes, decay routes, and charged-lepton mass-shape readouts are licensed downstream statuses, not primitive identity grounds. $\square$

Theorem 19.7 (Neutral weak matter certificate closure).

$$\text{ClosedMCCert}(C_{\text{neutral-weak-matter}}).$$

Proof. The certificate for $C_{\text{neutral-weak-matter}}$ is discharged in Section 8. It combines matter-representation standing, ChargeNeutral, weak-chiral response, and mass-response-slot standing. Flavor labels, oscillation baselines, PMNS entries, mass-splitting routes, and cosmological constraints are licensed route, mixing, measurement, or projection statuses. They are not primitive components of the neutrino carrier. $\square$

Theorem 19.8 (Mass-response-slot certificate closure).

$$\text{ClosedMCCert}(C_{\text{mass-response-slot}}).$$

Proof. The certificate for $C_{\text{mass-response-slot}}$ is discharged in Section 8. Higgs bridge/fixation and Yukawa compatibility supply the positive slot role; L1 Yukawa/mass equivalence supplies the negative overcount collapse. Numerical values and Yukawa coordinates remain skins, witnesses, or transport statuses. $\square$

Theorem 19.9 (Matter carrier package closure).

$$\text{ClosedMCert}(C_{\text{matter}}).$$

Proof. The five individual carrier certificates are closed in Theorems 19.4–19.8. The Matter Coverage Theorem, Theorem 19.3, discharges $\text{Coverage}_{D_{\text{matter}}^{N_g}}(C_{\text{matter}})$ by showing that every same-scope matter claim descends to the matter carrier package, receives a registered derivative or extension status, remains burdened, fails primitive promotion, or enters the hostile certificate branch. The hostile-residual clause does not neutralize that final branch: a certified same-scope primitive outside C_{matter} would falsify SM-3. Since no such closed hostile branch is supplied inside the fixed support stack and all open hostile assertions remain burdens rather than survivors, the package-level certificate closes. $\square$

Lemma 19.10 (Certificate-to-primitive-status deletion).

$$\text{ClosedMCert}(C_{\text{matter}}) \Rightarrow \text{Survprim}_{D_{\text{matter}}^{N_g}}(\text{Attr}(D_{\text{matter}}^{N_g}) \setminus C_{\text{matter}}) = 0.$$

Proof. Assume $y \in \text{Attr}(D_{\text{matter}}^{N_g}) \setminus C_{\text{matter}}$ is a same-scope primitive survivor while $\text{ClosedMCert}(C_{\text{matter}})$ holds. By Theorem 19.3, y either descends to C_{matter} , receives a registered status, remains burdened, fails primitive promotion, or enters $\text{HostileCertificateBranch}(y)$. Descent to C_{matter} contradicts $y \notin C_{\text{matter}}$. A registered status is derivative or extension status, not primitive survivor status. Burden status is review state, not survivor state. Failed primitive status is nonprimitive. If the hostile certificate branch closes, the hostile-residual clause says that the package certificate fails rather than absorbing the counterexample, contradicting $\text{ClosedMCert}(C_{\text{matter}})$. Thus no same-scope primitive survivor remains outside C_{matter} . $\square$

Theorem 19.11 (Scoped matter-sector primitive-status deletion).

$$\text{Survprim}_{D_{\text{matter}}^{N_g}}(\text{Attr}(D_{\text{matter}}^{N_g}) \setminus C_{\text{matter}}) = 0.$$

Proof. The proof has three steps. First, the five primitive matter carrier certificates close carrier-by-carrier in Theorems 19.4–19.8. Second, Theorem 19.3 covers every same-scope matter claim by carrier descent, registered derivative or extension status, burden, failure, or hostile-certificate exposure. Third, Lemma 19.10 deletes the residual primitive survivor branch: derivative status is not primitive, burden is not survivor, failure is nonprimitive, target update is not same-scope, and a closed hostile certificate falsifies the package rather than being absorbed. Hence the scoped residual primitive class is empty. $\square$

This theorem deletes only residual primitive-status survivors in the fixed same-scope matter arena. $\text{HostileResidual}_{D_{\text{matter}}^{N_g}}(C_{\text{matter}})$ does not convert a closed residual primitive into a registered status; it exposes the hostile branch. An uncertified primitive assertion remains burdened, and a certified same-scope primitive outside C_{matter} falsifies SM-3. The theorem does not delete lawful matter fields, particle labels, measured masses, tracks, jets, oscillations, EFT descriptions, BSM audits, or target-updating discoveries. It is not a closure of flavor/mixing coordinates, not a derivation of generation cardinality, not a numerical mass theorem, and not the SM-6 endpoint.

20 Minimal hostile matter reconstruction

20.1 Kernel-preserved hostile reconstruction

A hostile reconstruction has force only when it is a same-target reconstruction. It must preserve $D_{\text{matter}}^{N_g}$, preserve the comparison class, preserve reference standing, and then supply an independent primitive role not already carried or licensed by the closed matter package. Merely replacing the target, changing the admissibility rule, or appealing to a useful model surface is not a same-scope counterexample.

Criterion 20.1 (Faithful matter counterexample). *A faithful matter counterexample must show all of the following: (i) kernel-preserved same-scope membership in $D_{\text{matter}}^{N_g}$; (ii) primitive role occupancy; (iii) non-derivativeness from the certified matter carriers; (iv) nonmembership in the registered status atlas; and (v) target-preserving failure of the no-promotion theorem. If any item fails, the claim is a registered status, burden, failed primitive, or target update rather than a primitive survivor.*

A genuine local counterexample must be stronger than a new label, anomaly, track, oscillation result, mass value, or model surface. Let y_{matter}^* be a future matter claim such that:

1. $\text{MatterClaim}(y_{\text{matter}}^*)$;
2. $\text{SameScope}_{D_{\text{matter}}^{N_g}}(y_{\text{matter}}^*)$;
3. $y_{\text{matter}}^* \notin C_{\text{matter}}$;
4. it is not spinor notation, representation label, mass value, Yukawa coordinate, flavor route, detector route, hadron route, oscillation route, cosmological projection, EFT coordinate, model name, or generation label;
5. if it is generation-cardinality altering, it closes the role-cardinality certificate rather than merely duplicating field syntax;
6. it is not a target update;
7. it defeats derivative descent;
8. it supplies a closed matter carrier certificate.

Then

$$\text{Survprim}_{D_{\text{matter}}^{N_g}}(y_{\text{matter}}^* \setminus C_{\text{matter}})$$

and SM-3 fails. The classifier is not permitted to reclassify this object as target update unless the target-update criterion is satisfied, and it is not permitted to leave it as burden once the certificate closes. This hostile reconstruction is the local falsifier for SM-3.

21 Referee objection lock

Table 12: Referee objection lock

Objection	Response
The Standard Model defines matter by spinor fields.	Field representation is granted; primitive standing by spinor notation alone is denied.

Objection	Response
Quarks are particles.	Quark labels are preserved; isolated-particle primitive standing is denied under confinement-compatible color-bearing matter discipline.
Jets are evidence for quarks and gluons.	Yes. Jets are route/projection witnesses, not primitive isolated-quark or gluon standing.
Charged leptons are identified by their masses.	Masses are measurement witnesses or mass-response skins; they do not ground primitive matter identity.
Neutrino flavor is real because oscillations are observed.	Oscillations are preserved as boundary-to-boundary reconstruction routes; globally transported weak flavor identity is denied.
PMNS entries are physical.	PMNS coordinates are transition/mixing skins or measurement summaries; full flavor/mixing treatment belongs to SM-5.
Sterile neutrinos refute the closure.	Only if same-scope primitive neutrino standing outside C_ν closes. Otherwise sterile claims are route, extension, burden, target update, or failure.
This denies BSM fermions.	BSM fermion claims are audit-admissible; they are not protected from primitive-status classification.
This does not derive masses.	Correct. SM-3 is matter-carrier closure, not numerical mass closure.
You hid generation closure in SM-3.	No. SM-3 holds the family-cardinality envelope fixed and records generation-cardinality alteration as burden, failure, or target update unless role-cardinality closure is supplied.

22 Lean 4 formalization target and public audit reference

The Lean 4 release is used here as a paper-shaped, proof-checked audit and certificate surface for the SM manuscript sequence. It records the endpoint carriers, declared hypotheses, theorem crosswalk inventory, and certificate interfaces in Lean; verifies that the stated closure theorems typecheck without `sorry`, `admit`, live local `axiom` declarations, or `unsafe` code in the audit surface; and ties the formal release to the DOI-bearing manuscript bundle. This is a substantive support layer: it gives a typed theorem registry, explicit hypothesis boundaries, machine-checked closure over declared certificate fields, an audit trail across SM-1–SM-6 and the endpoint paper, reproducibility checks, and DOI/release alignment. It should not be described as an independent reconstruction of the full Standard Model argument from foundational physics.

The generation-cardinality endpoint should be read at the governance-to-endpoint interface, not as a standalone phenomenological derivation inside the Standard Model sector. The mechanized surface represents the pre-physical admissibility constraints under which a non-degenerate physical presentation is admissible: open-arena discipline, no-selector and no-presentation-dependence constraints, role-occupancy closure, and boundary handoff. Within that framework, generation cardinality is a formal closure condition rather than a fit to particle content or an inference from arbitrary presentation-dependent structure. The Lean release audits that certificate surface and its dependencies; it does not reconstruct every analytic or phenomenological argument in the SM

papers.

The public Lean-facing support package is the relevant SM ARC reference archive [34]: its README identifies the paper-shaped theorem surface, the locally received SM-1–SM-6 ARC layers, the generation-cardinality endpoint, the theorem-inventory layer, and the build/audit verifier. Its v0.1.0 release records the **GenerationCardinality** Lake target, carries SM-1 through SM-6 as local received ARC layers, preserves the generation-cardinality source artifacts, and reports build and focused axiom-audit checks over the endpoint anchors.

Lean audit item	Manuscript-facing role
Repository	somamaley-ux/AASC-Standard-Model-Carrier-Exhaustion-Generation-Cardinality, the public Lean 4 reference archive for the SM-1–SM-6 ARC carrier-exhaustion chain and the generation-cardinality endpoint.
Release	v0.1.0 - Standard Model Carrier Exhaustion / Generation Cardinality.
DOI	10.5281/zenodo.21219417.
Primary theorem surface	<code>SM3.lean</code> and its imported <code>SM3/...</code> files, read with <code>SM1.lean</code> , <code>SM2.lean</code> , <code>SM4.lean</code> – <code>SM6.lean</code> , and <code>GenerationCardinality.lean</code> as received support surfaces.
Axiom audit command	Repository verifier <code>scripts/check-generation-cardinality-audit.ps1</code> with the <code>-SkipLakeUpdate</code> option; the verifier builds the endpoint target, scans for live <code>axiom</code> , <code>sorry</code> , <code>admit</code> , and <code>unsafe</code> declarations, and runs focused axiom checks over endpoint anchors.
Build target	<code>GenerationCardinality</code> , with aggregate imports through <code>SM1.lean</code> – <code>SM6.lean</code> and <code>GenerationCardinality.lean</code> .
Audit meaning	The public release machine-checks the governance-to-endpoint certificate surface over declared carriers, received ARC layers, and explicit manuscript hypotheses. Once the pre-physical admissibility layer is accepted, the endpoint predicate is a formal closure condition, not a phenomenological fit or downstream presentation selector.
Certificate/interface value	The layer is not merely bookkeeping: it supplies a typed theorem registry, explicit hypothesis boundaries, machine-checked closure over declared certificate fields, an SM-1–SM-6 audit crosswalk, reproducibility checks, and DOI/release alignment.
Boundary	The cardinality endpoint is not a standalone SM-sector derivation from particle dynamics; it is a governance-level endpoint that makes explicit the admissibility, no-selector, open-arena, role-occupancy, and presentation-independence constraints under which later SM interpretations are meaningful. The Lean release audits this certificate/interface boundary and its dependencies; it does not reconstruct every analytic or phenomenological argument of SM-1–SM-6.

A manuscript-faithful Lean route for SM-3 represents:

1. fixed scopes and submitted-claim or candidate-record sets as explicit carriers;
2. carrier packages, registered-status packages, and received sector packages as typed structures rather than implicit background metaphysics;
3. same-scope and target-update predicates as explicit boundary conditions;
4. derivative-skin, representative, route, projection, transport, burden, failure, and extension statuses as typed predicates or classifiers;
5. certificate closure and no-promotion principles as theorems over declared hypotheses, not as global axioms;
6. finite branch exhaustions for the relevant same-scope arena;
7. the handoff to SM-6, or from SM-6 to the generation-cardinality endpoint, as an explicit typed support relation.

For SM-3, the paper-specific audit surface is the matter-sector package: matter-representation carrier, color-bearing quark standing, colorless charged-lepton standing, neutral weak neutrino standing, flavor-route interface status, mass/Yukawa and mass-response slot discipline, generation-label non-promotion, sterile-neutrino branch typing, and scoped matter primitive-status deletion.

A manuscript-faithful formalization over the declared carriers and hypotheses requires no custom axiom at a load-bearing node. If a proposed audit file relies on `axiom`, `sorry`, `admit`, `unsafe`, or an opaque proof-carrying placeholder at such a node, it is not counted as formal support for this manuscript. Conversely, a clean axiom audit is not a stand-alone validation of the physical hypotheses encoded as record fields; it is a proof-checked governance-to-endpoint certificate boundary whose value is typed traceability, explicit assumptions, reproducible build discipline, and DOI-aligned auditability.

23 Handoff to later papers

23.1 Handoff to SM-4

SM-3 hands forward the mass/Yukawa bookkeeping boundary, no primitive mass-value theorem, no primitive Yukawa-entry theorem, and charge-response status for charged matter. SM-4 handles charges, couplings, renormalization, running values, matching conditions, and transport/calibration ledgers.

23.2 Handoff to SM-5

SM-3 hands forward flavor-label non-primitivity, neutrino boundary/flavor route discipline, PMNS coordinate as mixing skin, and generation-label burden/fixed-boundary discipline. SM-5 handles CKM, PMNS, CP phases, basis choices, generation cardinality, and flavor-transition carrier closure.

23.3 Handoff to SM-6

SM-3 hands forward C_{matter} , the matter skin atlas, quark, charged-lepton, and neutrino certificates, the mass/Yukawa non-promotion theorem, detector/route/projection atlas, BSM/EFT matter branch certificates, matter prediction register M0–M10, and the scoped matter primitive-status deletion theorem. SM-6 may use this as the matter-sector component of

$$C_{SM} = C_{\text{gauge}} \cup C_{\text{matter}} \cup C_{\text{transport}} \cup C_{\text{flavor}} \cup C_H \cup C_{\text{projection}}.$$

24 Conclusion

The kernel-status strengthening is part of the theorem, not a commentary layer. The closure claimed here is primitive-status closure under the minimal non-degeneracy conditions required for matter claims to remain the same standing-bearing physical claims across measurement, prediction, comparison, and attempted falsification.

Within the fixed Standard Model matter comparison arena $D_{\text{matter}}^{N_g}$, no residual same-scope primitive matter-sector attribute survives outside the closed matter carrier network. Quark language is licensed by color-bearing representation-indexed matter standing under confinement-compatible discipline. Charged-lepton language is licensed by colorless charged matter-response standing. Neutrino language is licensed by neutral weak matter standing, with flavor and oscillation preserved as boundary/route, mixing, measurement, or projection statuses. Spinor notation, representation labels, mass values, Yukawa entries, flavor names, generation labels, detector tracks, hadron membership, oscillation baselines, cosmological bounds, EFT coefficients, and BSM matter labels remain representatives, witnesses, transport ledgers, routes, projections, extensions, target updates, burdens, failures, or genuine falsifiers according to the stated certificate conditions.

SM-3 therefore proves

$$\text{Survprim}_{D_{\text{matter}}^{N_g}} (\text{Attr}(D_{\text{matter}}^{N_g}) \setminus C_{\text{matter}}) = 0.$$

This is primitive matter closure with skin openness. It is not empirical no-new-physics, not numerical parameter closure, not flavor/mixing closure, not generation-cardinality closure, and not the global Standard Model endpoint. It is the matter-sector component required for the eventual SM-6 carrier-exhaustion synthesis.

A Expanded dependency ledger

Dependency	Imported payload	Carrier affected	Nonclaim boundary
SM-1	Master classifier and no-promotion theorem	all statuses	Does not prove matter carriers by itself.
SM-2	Gauge/color/weak carrier package	quark, charged lepton, neutrino	Does not prove matter side alone.
Standalone SM spine	Structural tuple and parameter-status boundary	full arena	Does not derive full empirical parameter table.
Realized physical interior	Fields and particles as physically realized roles	representation carrier	Does not turn field symbols into primitive matter identity.
A4 representation response	Pre-linear representation and charge-witness ledgers	matter-rep	Does not assume vector spaces or representation theory.
PSM3 representation algebra	Witness-preserving algebraic specialization	matter-rep skins	Does not derive Standard Model multiplets or charges.

SR3 representation rigidity	Response role compression and multiplicity discipline	matter-rep, generation labels	Does not close all generation questions.
Charge lattice and L1 charge	Charge normalization discipline	charged and neutral predicates	Does not derive numerical gauge couplings.
L1 anomaly/Yukawa	Yukawa realizability under chiral/anomaly discipline	chiral and BSM guards	Does not derive Yukawa eigenvalues.
L1 Yukawa/mass	Mass/Yukawa bookkeeping equivalence	mass/Yukawa skins	Negative overcount collapse, not full mass-slot closure alone.
MGN4	Yukawa compatibility and parameter-survival classification	mass-response slot	Does not close numerical mass spectrum.
Generation-count support	Fixed-boundary family envelope	generation label discipline	Does not classify every fourth-generation model by name.
Charged-lepton support	Mass-shape status and readout discipline	charged-lepton skins	Does not make SM-3 a charged-lepton numerical closure paper.
Neutrino/AOT support	Oscillation, branch, matter-response, observable transport	neutrino route/projection skins	Does not close PMNS or absolute mass scale.

B Full matter skin atlas

Skin	Dependence test	Licensed status
Spinor field	Field/representation notation	Licensed representative.
Vector space	Algebraic realization of response ledger	Representative realization.
Basis	Representative choice	Skin unless witness-changing.
Chiral label	Witnessed handed response required	Chiral response status or skin.
Representation label	Raw label does not self-ground	Response representative or failure.
Dimension	Witness-bearing degree required	Multiplicity/dimension status or skin.
Generation label	Fixed-boundary envelope required	Family-index status or burden.
Generation-cardinality alteration	Changes family envelope unless certified	Burden, failure, target update, or SM-5/SM-6 certificate.
Mass value	Measurement/slot dependent	Mass-response witness.
Yukawa coordinate	Higgs/mass bookkeeping dependent	Mass-response or transport status.

Flavor name	Boundary/route dependent	Route or transition label.
Detector track	Measurement apparatus dependent	Measurement/projection witness.
Hadron membership	Confinement route dependent	Route status.
Parton distribution	Projection and scheme dependent	Projection/transport route.
Jet	Detector and hadronization dependent	Route/projection witness.
Oscillation baseline	Source-to-detector route dependent	Route/measurement status.
PMNS entry	Basis/phase/coordinate dependent	Mixing-coordinate skin.
Cosmological bound	Projection/regime dependent	Projection status.
Sterile-neutrino label	Model surface, route, or hidden-sector dependence	Branch-certificate input.
Vector-like fermion label	Field/model surface plus chiral/anomaly tests	Extension, burden, update, or failure.
Leptoquark label	Simultaneous color/charge/matter role tests	Extension, route, projection, burden, update, or failure.
Compositeness surface	Reconstruction or target-change dependence	Route, projection, burden, update, or failure.
EFT coefficient	Operator-basis/matching dependent	Projection/transport status.
BSM model name	Model surface dependent	Branch-certificate input.

C Detailed quark certificate

Field	Certificate content
Submitted object	Quark field, flavor label, color label, current mass, constituent mass, parton route, hadron membership, isolated-quark claim.
Fixed target	$D_{\text{matter}}^{N_g}$ with representation response, SM-2 color transport, anomaly/vacuum discipline, and mass-response boundary fixed.
Carrier candidate	$C_{\text{color-bearing-matter}} = C_{\text{matter-rep}} \wedge C_{\text{color-transport}} \wedge \text{ConfCompat}$.
Licensed skins	Spinor field notation, flavor labels, current/constituent masses, hadron route language, parton distributions, jets.
False primitive deleted	Isolated-particle skin, mass label, flavor label, and hadron membership as primitive quark identity.
Falsifier	Free quark with same-scope primitive SM standing outside $C_{\text{color-bearing-matter}}$, with target update and derivative descent excluded.

Endpoint line	The quark primitive-status theorem is closed locally by color-bearing matter standing.
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D Detailed charged-lepton certificate

Field	Certificate content
Submitted object	Electron, muon, tau labels; spinor fields; pole masses; tracks; lifetimes; decay routes; Yukawa coordinates; family labels.
Fixed target	$D_{\text{matter}}^{N_g}$ with representation response, charge response, weak-chiral response, Higgs/mass slot, anomaly/Yukawa compatibility, and family envelope fixed.
Carrier candidate	$C_{\text{colorless-charged-matter}} = C_{\text{matter-rep}} \wedge \text{ColorNull} \wedge C_{\text{charge-response}} \wedge C_{\text{weak-chiral}} \wedge C_{\text{mass-response-slot}}$.
Licensed skins	Field labels, electric charge values, pole masses, decay routes, detector tracks, mass ratios, Yukawa entries.
False primitive deleted	Mass value, track, family label, or Yukawa entry as primitive charged-lepton identity.
Falsifier	Same-scope primitive charged-lepton carrier outside $C_{\text{colorless-charged-matter}}$, with certificate closed.
Endpoint line	The charged-lepton primitive-status theorem is closed locally by colorless charged matter standing.

E Detailed neutrino certificate

Field	Certificate content
Submitted object	Neutrino field label, weak flavor label, mass eigenstate label, oscillation baseline, PMNS entry, mass splitting, cosmological bound, sterile-neutrino surface.
Fixed target	$D_{\text{matter}}^{N_g}$ with representation response, charge-neutral status, weak-chiral response, mass-response slot, and observable transport discipline fixed.
Carrier candidate	$C_\nu = C_{\text{neutral-weak-matter}}$, with $C_{\text{matter-rep}}$, ChargeNeutral , $C_{\text{weak-chiral}}$, and $C_{\text{mass-response-slot}}$.
Licensed skins	Neutrino labels, flavor names, PMNS coordinates, oscillation baselines, mass splittings, cosmological bounds, missing-energy displays.
False primitive deleted	Flavor label, oscillation route, PMNS coordinate, cosmological bound, or sterile label as primitive neutrino identity.
Falsifier	Same-scope primitive neutrino carrier outside C_ν , with target update and derivative descent excluded.

Endpoint line	The neutrino primitive-status theorem is closed locally by neutral weak matter standing plus licensed flavor route/interface status.
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F Mass/Yukawa certificate

Field	Certificate content
Submitted object	Fermion mass value, pole mass, current mass, constituent mass, Yukawa coordinate, Higgs-Yukawa expression, low-energy mass parameter.
Fixed target	$D_{\text{matter}}^{N_g}$ with Higgs bridge/fixation, mass-response slot, representation response, and weak/charge/color carrier roles fixed.
Carrier candidate	$C_{\text{mass-response-slot}}$.
Licensed skins	m_f, y_f, v , pole/current/constituent mass displays, threshold and propagator routes.
False primitive deleted	Mass or Yukawa coordinate as primitive matter identity.
Falsifier	A mass/Yukawa coordinate supplying primitive matter identity outside mass-response-slot descent under fixed $D_{\text{matter}}^{N_g}$.
Endpoint line	Mass/Yukawa labels are licensed slot or transport/measurement statuses, not primitive carriers.

G Standard physics orientation: ordinary usage being reclassified

The internal AASC sources carry the proof burden. Standard physics references fix the conventional language being reclassified and prevent a simpler misreading: SM-3 is not discarding standard particle-physics practice. It is assigning primitive, representative, measurement, transport, route, projection, burden, failure, or target-update status to the familiar objects of that practice.

G.1 SM fermion fields and representations

Standard Model matter is usually presented through chiral fermion fields and representation assignments. Left-handed quark and lepton representatives, right-handed singlet representatives, conjugate fields, multiplets, and spinor notation remain the ordinary calculational language. SM-3 preserves that language as representative and algebraic realization language. The representation-response carrier is not a chosen spinor basis, multiplet label, or finite-dimensional vector-space display; those displays are licensed only after they preserve the response ledger [29, 30, 31].

G.2 Quarks, confinement, partons, and jets

Quark, parton, hadron, and jet language remains central to phenomenology. Parton-model descriptions, hadron membership, scattering channels, and jets are powerful evidence and route language for color-bearing matter response. SM-3 does not demote their empirical role. It denies only isolated-quark primitive standing by label, current/constituent mass display, parton route, or jet topology alone. A same-scope free-quark primitive would have to close the hostile certificate outside the color-bearing matter carrier.

G.3 Charged leptons and pole masses

Charged-lepton pole masses, lifetimes, decay channels, and detector tracks remain empirical witnesses and readout statuses. The ordinary electron, muon, and tau labels are preserved. The primitive-status point is narrower: no pole value, track, lifetime, family label, or Yukawa coordinate grounds charged-lepton identity without the colorless charged matter-response carrier.

G.4 Neutrino oscillation and PMNS language

Neutrino oscillation and PMNS language remains valid route and transition-coordinate language. Oscillation baselines, mass-splitting fits, matter effects, and PMNS coordinates are empirical and phenomenological surfaces of neutral weak matter standing. SM-3 places them in route, measurement, projection, or mixing status. Full transition-coordinate closure belongs to SM-5, and absolute mass scale, ordering, and Dirac/Majorana status remain downstream or burdened.

G.5 EFT and BSM fermion model building

EFT operators and new fermion labels remain admissible audit inputs [32, 33]. Operator coefficients, vector-like fermion labels, sterile-neutrino surfaces, leptoquark labels, compositeness proposals, and hidden-sector fermion models may be useful probes. SM-3 asks the primitive-status question: do they descend to existing carriers, function as route/projection/transport language, change the target, remain burdened, fail primitive promotion, or close the hostile same-scope certificate? They are not protected from classification by being model-building vocabulary.

H Theorem inventory

Theorem	SM-3 status	Handoff
Matter skin non-promotion	Closed local theorem	Used by SM-6.
ColorNull predicate definition	Definition/criterion	Used by charged-lepton certificate.
ChargeNeutral predicate definition	Definition/criterion	Used by neutrino certificate.
ConfCompat definition	Criterion	Used by quark certificate.
Spinor field non-primitivity	Closed local corollary	Used by SM-6.
Representation-label non-primitivity	Closed local corollary	Used by SM-5/SM-6.
Quark color-bearing matter theorem	Closed local theorem	Used by SM-6.
Charged-lepton colorless charged matter theorem	Closed local theorem	Used by SM-6.
Neutrino neutral weak matter theorem	Closed local theorem	Used by SM-5/SM-6.
Flavor route non-carrier theorem	Closed local corollary	Used by SM-5.

Mass/Yukawa promotion	negative no-	Closed local theorem	Boundary to SM-4/SM-5.
Mass-response-slot rier theorem	positive car-	Closed local theorem	Boundary to SM-4/SM-6.
Generation-label promotion	non-	Closed local corollary	Handoff to SM-5.
Generation-cardinality branch	burden	Branch criterion	Handoff to SM-5/SM-6.
Sterile-neutrino certificate	full branch cer-	Branch logic closed	Named models require instantiation.
Carrier-by-carrier closed certificate theorem	closed certifi-	Closed local theorem	Used by SM-6.
Matter BSM branch certificate	branch certificate	Branch logic closed	Named models require instantiation.
Scoped matter deletion		Closed local theorem	SM-6 receives.
Global SM endpoint		Not claimed	SM-6 target.

I SM-6 release certificate

SM-6 receives from SM-3:

1. C_{matter} , the matter carrier network;
2. the matter skin atlas;
3. quark, charged-lepton, and neutrino standing certificates;
4. the mass/Yukawa non-promotion theorem;
5. detector, route, and projection atlas;
6. BSM/EFT matter branch certificates;
7. the matter prediction register M0–M10;
8. the hostile matter reconstruction y_{matter}^* ;
9. the scoped primitive-status deletion theorem

$$\text{Survprim}_{D_{\text{matter}}^{N_g}} (\text{Attr}(D_{\text{matter}}^{N_g}) \setminus C_{\text{matter}}) = 0.$$

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